Python Solutions: Maths Questions 1-20

Each solution takes its values from the user with input() and prints the result. `import math` is required.

```
import math
```

1) Area of a Rectangle

Formula: Area = length x width

```
length = float(input("Enter length: "))
width = float(input("Enter width: "))
print("Area =", length * width)
```

2) Circumference of a Circle

Formula: Circumference = 2pi r

```
r = float(input("Enter radius: "))
print("Circumference =", 2 * math.pi * r)
```

3) Simple Interest

Formula: SI = Principal x Rate x Time

```
p = float(input("Enter principal: "))
rate = float(input("Enter rate (as a decimal, e.g. 0.05): "))
time = float(input("Enter time (years): "))
print("Simple Interest =", p * rate * time)
```

4) Speed of an Object

Formula: Speed = Distance / Time

```
distance = float(input("Enter distance: "))
time = float(input("Enter time: "))
if time == 0:
    print("Time cannot be zero.")
else:
    print("Speed =", distance / time)
```

5) BMI Calculator

Formula: BMI = Weight(kg) / Height(m)^2

```
weight = float(input("Enter weight (kg): "))
height = float(input("Enter height (m): "))
if height == 0:
    print("Height cannot be zero.")
else:
    print("BMI =", weight / height ** 2)
```

6) Force Using Newton's Second Law

Formula: F = m x a

```
m = float(input("Enter mass (kg): "))
a = float(input("Enter acceleration (m/s^2): "))
print("Force =", m * a, "N")
```

7) Compound Interest

Formula: A = P(1 + r/n)^{nt}

```
p = float(input("Enter principal (P): "))
r = float(input("Enter annual rate as decimal (r): "))
n = float(input("Enter compounds per year (n): "))
t = float(input("Enter time in years (t): "))
a = p * (1 + r / n) ** (n * t)
print("Total amount A =", a)
print("Compound interest =", a - p)
```

8) Perimeter of a Triangle

Formula: Perimeter = a + b + c

```
a = float(input("Enter side a: "))
b = float(input("Enter side b: "))
c = float(input("Enter side c: "))
print("Perimeter =", a + b + c)
```

9) Volume of a Sphere

Formula: V = (4/3) x pi x r^3

```
r = float(input("Enter radius: "))
print("Volume =", (4 / 3) * math.pi * r ** 3)
```

10) Kinetic Energy

Formula: KE = (1/2) m v^2

```
m = float(input("Enter mass (kg): "))
v = float(input("Enter velocity (m/s): "))
print("Kinetic Energy =", 0.5 * m * v ** 2, "J")
```

11) Quadratic Equation Roots

Formula: x = (-b +/- sqrt(b^2 - 4ac)) / 2a

```
a = float(input("Enter a: "))
b = float(input("Enter b: "))
c = float(input("Enter c: "))
if a == 0:
    print("a cannot be zero for a quadratic equation.")
    return
d = b ** 2 - 4 * a * c
if d >= 0:
    x1 = (-b + math.sqrt(d)) / (2 * a)
    x2 = (-b - math.sqrt(d)) / (2 * a)
else:
    real = -b / (2 * a)
    imag = math.sqrt(-d) / (2 * a)
    x1 = complex(real, imag)
    x2 = complex(real, -imag)
print("Roots:", x1, "and", x2)
```

12) Temperature Conversion

Formula: F = C x 9/5 + 32

```
c = float(input("Enter temperature in Celsius: "))
print("Fahrenheit =", c * 9 / 5 + 32)
```

13) Gravitational Force

Formula: F = G m1 m2 / r^2

```
G = 6.674e-11
m1 = float(input("Enter mass m1 (kg): "))
m2 = float(input("Enter mass m2 (kg): "))
r = float(input("Enter distance r (m): "))
if r == 0:
    print("Distance cannot be zero.")
else:
    print("Gravitational Force =", G * m1 * m2 / r ** 2, "N")
```

14) Volume of a Cylinder

Formula: $V = \pi r^2 h$

```
r = float(input("Enter radius: "))
h = float(input("Enter height: "))
print("Volume =", math.pi * r ** 2 * h)
```

15) Pressure

Formula: P = F / A

```
f = float(input("Enter force: "))
a = float(input("Enter area: "))
if a == 0:
    print("Area cannot be zero.")
else:
    print("Pressure =", f / a)
```

16) Electric Power

Formula: P = V x I

```
v = float(input("Enter voltage (V): "))
i = float(input("Enter current (A): "))
print("Power =", v * i, "W")
```

17) Perimeter of a Circle (Circumference)

Formula: Perimeter = 2 pi r

```
r = float(input("Enter radius: "))
print("Perimeter =", 2 * math.pi * r)
```

18) Future Value in Savings

Formula: $FV = PV (1 + r)^t$

```
pv = float(input("Enter present value (PV): "))
r = float(input("Enter annual rate as decimal (r): "))
t = float(input("Enter time in years (t): "))
print("Future Value =", pv * (1 + r) ** t)
```

19) Work Done by a Force

Formula: W = f x d x cos(theta)

```
f = float(input("Enter force: "))
d = float(input("Enter distance: "))
theta = float(input("Enter angle (degrees): "))
print("Work done =", f * d * math.cos(math.radians(theta)), "J")
```

20) Heat Transfer

Formula: Q = m c (delta T)

```
m = float(input("Enter mass: "))
c = float(input("Enter specific heat capacity: "))
dt = float(input("Enter temperature change (delta T): "))
print("Heat transferred =", m * c * dt)
```